

Section 11.4: Comparison Tests for Positive Series

Let $\sum_{n=1}^{\infty} a_n$ be a positive series (i.e., $a_n \geq 0$ for all n) and let $s_n = \sum_{k=1}^n a_k$ be the

n -th partial sum. Then $\{s_n\}$ is an increasing sequence and hence it is bounded below by a_1 . By the Monotonic Sequence Theorem, $\{s_n\}$ is convergent if and only if $\{s_n\}$ is bounded above. Therefore, we have

$$\sum_{n=1}^{\infty} a_n \text{ is convergent} \iff \{s_n\} \text{ is bounded above}$$

and

$$\sum_{n=1}^{\infty} a_n \text{ is divergent} \iff \{s_n\} \text{ is unbounded above.}$$

We will write $\sum_{n=1}^{\infty} a_n < \infty$ when the positive series is convergent and write $\sum_{n=1}^{\infty} a_n = \infty$

when the positive series is divergent.

The Direct Comparison Test. Suppose $0 \leq a_n \leq b_n$ for all $n \geq n_0$, where n_0 is a positive integer.

(i) If $\sum_{n=1}^{\infty} b_n < \infty$, then $\sum_{n=1}^{\infty} a_n < \infty$.

(ii) If $\sum_{n=1}^{\infty} a_n = \infty$, then $\sum_{n=1}^{\infty} b_n = \infty$.

Remark. We should know the convergence/divergence of some “standard” positive series so that many other series can be compared with them.

Example 1. Determine the convergence or divergence of the following series.

$$(i) \quad \sum_{n=1}^{\infty} \frac{5}{2n^2 + 4n + 3}.$$

The series is convergent, since for all n , we have

$$\frac{5}{2n^2 + 4n + 3} \leq \frac{5}{2n^2} = \frac{5}{2} \frac{1}{n^2}$$

and $\sum_{n=1}^{\infty} \frac{5}{2} \frac{1}{n^2} = \frac{5}{2} \sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent (the p -series with $p = 2 > 1$).

$$(ii) \quad \sum_{n=1}^{\infty} \frac{\ln n}{n}.$$

The series is divergent, since $\frac{\ln n}{n} \geq \frac{1}{n}$ for all $n \geq 3$ and $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

$$(iii) \quad \sum_{n=1}^{\infty} \frac{n+5}{(n^7 + n^2)^{1/3}}.$$

It is convergent, since $\frac{n+5}{(n^7 + n^2)^{1/3}} \leq \frac{2n}{n^{7/3}} = \frac{2}{n^{4/3}}$ for all $n \geq 5$ and $\sum_{n=1}^{\infty} \frac{2}{n^{4/3}}$ is convergent.

$$(iv) \quad \sum_{n=1}^{\infty} \frac{n+5}{(n^7 + n^2)^{1/4}}.$$

It is divergent, since

$$\frac{n+5}{(n^7 + n^2)^{1/4}} \geq \frac{n}{(2n^7)^{1/4}} = \frac{1}{\sqrt[4]{2}} \frac{1}{n^{3/4}} \quad \text{for all } n$$

and $\sum_{n=1}^{\infty} \frac{1}{\sqrt[4]{2}} \frac{1}{n^{3/4}} = \frac{1}{\sqrt[4]{2}} \sum_{n=1}^{\infty} \frac{1}{n^{3/4}}$ is divergent. ■

By the Direct Comparison Test, we know that if $a_n \sim b_n$ for n large enough, then the two series have the same convergence/divergence. However, sometimes a direct comparison is not convenient. In this case, we can consider the limit of the ratio $\frac{a_n}{b_n}$.

The Limit Comparison Test. Suppose that $a_n > 0$ and $b_n > 0$ for all n . If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c > 0,$$

then either both $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ are convergent or both are divergent.

Remark. The Test fails when $c = 0$, or $c = \infty$, or the limit does not exist.

Why the Limit Comparison Test holds. When n is large enough, we have

$$\frac{1}{2}c \leq \frac{a_n}{b_n} \leq \frac{3}{2}c;$$

that is, $\frac{c}{2}b_n \leq a_n \leq \frac{3c}{2}b_n$. Therefore, by the Direct Comparison Test, $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ have the same convergence/divergence.

If we look again the series in Example 1(i), (iii) and (iv), we can quickly conclude the convergence/divergence by considering the limit of the ratio of a_n and

$$b_n = \frac{1}{n^2}, \quad \frac{1}{n^{4/3}}, \quad \frac{1}{n^{3/4}}, \quad \text{respectively.}$$

Example 2. Determine the convergence or divergence of the following series.

$$(1) \quad \sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{n^2 - 1}{3n^4 + 2n + 5}.$$

Take $b_n = \frac{1}{n^2}$. Then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n^4 - n^2}{3n^4 + 2n + 5} = \frac{1}{3} > 0$. Therefore, the series is convergent.

$$(2) \quad \sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{2^{1/n}}{n}.$$

Take $b_n = \frac{1}{n}$. Then $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} 2^{1/n} = 1 > 0$. Therefore, the series is divergent.

$$(3) \quad \sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{n + 4^n}{2n + 5^n}.$$

By L'Hospital's Rule, for $a > 1$, $\lim_{x \rightarrow \infty} \frac{x}{a^x} = \lim_{x \rightarrow \infty} \frac{1}{a^x \ln a} = 0$. Take $b_n = \frac{4^n}{5^n} = \left(\frac{4}{5}\right)^n$.

Then

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{1 + \frac{n}{4^n}}{1 + \frac{2n}{5^n}} = 1 > 0.$$

Therefore, the series is convergent. ■

Section 11.5: Alternating Series and Absolute Convergence

We know that the harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \cdots$ is divergent.

However, the series $1 - \frac{1}{2} + \frac{1}{3} - \cdots$ is convergent:

$$s_4 = s_2 + \left(\frac{1}{3} - \frac{1}{4}\right) > s_2,$$

$$s_{2n} = s_{2n-2} + \left(\frac{1}{2n-1} - \frac{1}{2n}\right) > s_{2n-2}.$$

Thus we have $s_2 \leq s_4 \leq s_6 \leq \cdots$. Also, since

$$s_{2n} = 1 - \left(\frac{1}{2} - \frac{1}{3}\right) - \left(\frac{1}{4} - \frac{1}{5}\right) - \cdots - \left(\frac{1}{2n-2} - \frac{1}{2n-1}\right) - \frac{1}{2n} < 1,$$

$\{s_{2n}\}$ is increasing and bounded above. Therefore, $\lim_{n \rightarrow \infty} s_{2n} = s$ exists. Now we have

$$s_{2n+1} = s_{2n} + \frac{1}{2n+1} \rightarrow s.$$

Therefore, $\{s_n\}$ is convergent.

Alternating Series Test. Let $\{b_n\}$ be a decreasing sequence such that $\lim_{n \rightarrow \infty} b_n = 0$;

that is,

$$b_1 \geq b_2 \geq b_3 \geq \cdots \rightarrow 0.$$

Then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ and hence $\sum_{n=1}^{\infty} (-1)^n b_n$ are convergent.

Remark. The Alternating Series Test holds if the alternating series satisfies

$$b_k \geq b_{k+1} \geq b_{k+2} \geq \cdots \rightarrow 0,$$

where k is some positive integer.

Example 3. Determine the convergence or divergence of the following series.

$$(1) \quad \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n}.$$

In this alternating series, $b_n = \frac{1}{2n} \downarrow 0$. Therefore, the series is convergent.

$$(2) \quad \sum_{n=1}^{\infty} (-1)^{n+1} \frac{n^2}{n^3 + 1} = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{n^2}{n^3 + 1}.$$

In this alternating series, $b_n = \frac{n^2}{n^3 + 1} \downarrow 0$, since $f(x) = \frac{x^2}{x^3 + 1}$ is decreasing on $[2, \infty)$

$\left(f'(x) = \frac{x(2 - x^3)}{(x^3 + 1)^2} < 0\right)$ and $f(x) \rightarrow 0$ as $x \rightarrow \infty$. Therefore, the series is convergent.

$$(3) \quad \sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{2n}{3n - 1}.$$

In this alternating series, the sequence $\{b_n\}$ is decreasing but $b_n \not\rightarrow 0$. In fact,

$$|a_n| = \frac{2n}{3n - 1} = \frac{2}{3 - \frac{1}{n}} \rightarrow \frac{2}{3} \neq 0.$$

Therefore, by the Test for Divergence, the given series is divergent.

$$(4) \quad \sum_{n=1}^{\infty} a_n = 1 - \frac{1}{2^2} + \frac{1}{3} - \frac{1}{4^2} + \frac{1}{5} - \frac{1}{6^2} + \cdots.$$

$$\text{We have } b_n = \begin{cases} \frac{1}{n} & \text{if } n \text{ is odd} \\ \frac{1}{n^2} & \text{if } n \text{ is even} \end{cases}.$$

Thus $b_n \rightarrow 0$ but $\{b_n\}$ is not decreasing. Now $\sum_{n=1}^{\infty} a_n$ is divergent, since

$$\sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} \frac{1}{n^2} > 1 + \frac{1}{3} + \frac{1}{5} + \cdots = \sum_{n=1}^{\infty} \frac{1}{2n - 1}$$

and $\sum_{n=1}^{\infty} \frac{1}{2n - 1}$ is divergent (since $\frac{1}{2n - 1} > \frac{1}{2n}$ and $\sum_{n=1}^{\infty} \frac{1}{2n}$ is divergent). ■

Remark. The last two series in Example 3 show that an alternating series $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ can be divergent if either $b_n \not\rightarrow 0$ or $\{b_n\}$ is not decreasing.

Definition. A series $\sum_{n=1}^{\infty} a_n$ is called absolutely convergent if $\sum_{n=1}^{\infty} |a_n| < \infty$, and is called conditionally convergent if $\sum_{n=1}^{\infty} a_n$ is convergent but not absolutely convergent.

The Relation between Convergence and Absolute Convergence. We have

$$\text{absolute convergence} \implies \text{convergence},$$

but the converse is not true:

Suppose $\sum_{n=1}^{\infty} |a_n| < \infty$. Let $b_n = a_n + |a_n|$. Then $0 \leq b_n \leq 2|a_n|$ and hence $\sum_{n=1}^{\infty} b_n$ is convergent. Therefore, $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} (b_n - |a_n|) = \sum_{n=1}^{\infty} b_n - \sum_{n=1}^{\infty} |a_n|$ is convergent.

Note that $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n}$ is convergent but not absolutely convergent. Therefore,

$$\text{convergence} \not\Rightarrow \text{absolute convergence}.$$

Example 4. We consider the alternating p -series $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^p}$.

It is known that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ convergent exactly when $p > 1$. Therefore, $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^p}$ is absolutely convergent exactly when $p > 1$.

For $0 < p \leq 1$, by the Alternating Series Test, $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^p}$ is conditionally convergent since $\frac{1}{n^p} \downarrow 0$ as $n \rightarrow \infty$.

For $p < 0$, by the Test for Divergence, $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^p}$ is divergent as $\left| (-1)^n \frac{1}{n^p} \right| \not\rightarrow 0$. ■

Section 11.6: The Ratio and Root Tests

For a positive series $\sum_{n=1}^{\infty} a_n$, in Section 11.4, we discussed the Limit Comparison Test,

where we compare $\sum_{n=1}^{\infty} a_n$ with another positive series $\sum_{n=1}^{\infty} b_n$ by considering $\lim_{n \rightarrow \infty} \frac{a_n}{b_n}$.

In this section, we discuss two tests for absolute convergence (and hence for convergence of positive series). They are also tests of comparison (comparing with itself rather than comparing with another series).

The Ratio Test. Let $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$. Then $0 \leq L \leq \infty$.

(i) If $L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent (and hence convergent).

(ii) If $L > 1$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

(iii) If $L = 1$, then there is no conclusion on $\sum_{n=1}^{\infty} a_n$ and the Ratio test fails.

Why the Ratio Test holds. For large n , we have $\left| \frac{a_{n+1}}{a_n} \right| \sim L$ and thus

$$|a_{n+1}| \sim L |a_n| \sim L^2 |a_{n-1}| \sim L^3 |a_{n-2}| \sim \dots$$

Therefore, we can compare the series $\sum_{n=1}^{\infty} a_n$ with the geometric series $\sum_{n=1}^{\infty} L^n$. In this way, we get parts (i) and (ii) in the test.

For part (iii), we can look at the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ and $\sum_{n=1}^{\infty} \frac{1}{n}$, where $L = 1$ for both series.

However, now one series is convergent but the other one is divergent.

Example 5. Determine the absolute convergence or divergence of the following series.

$$(1) \sum_{n=1}^{\infty} (-1)^n \frac{n^3}{3^n}.$$

Now $\left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{3} \left(1 + \frac{1}{n} \right)^3 \rightarrow \frac{1}{3} < 1$. Therefore, the series is absolutely convergent.

$$(2) \sum_{n=1}^{\infty} \frac{n!}{1000^n}.$$

Now $\left| \frac{a_{n+1}}{a_n} \right| = \frac{n+1}{1000} \rightarrow \infty > 1$. Therefore, the series is divergent.

$$(3) \sum_{n=1}^{\infty} \frac{n^n}{(n!)^2}.$$

Now $\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^{n+1} n! n!}{(n+1)! (n+1)! n^n} = \left(1 + \frac{1}{n} \right)^n \frac{1}{n+1} \rightarrow e \cdot 0 = 0 < 1$. Therefore,

the series is convergent.

Now we turn to the Root Test for absolute convergence.

The Root Test. Let $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$. Then $0 \leq L \leq \infty$.

(i) If $L < 1$, then the series $\sum_{n=1}^{\infty} a_n$ is absolutely convergent (and hence convergent).

(ii) If $L > 1$, then the series $\sum_{n=1}^{\infty} a_n$ is divergent.

(iii) If $L = 1$, then there is no conclusion on $\sum_{n=1}^{\infty} a_n$ and the Root test fails.

Why the Root Test holds. For large n , we have $|a_n|^{1/n} \sim L$ and thus $|a_n| \sim L^n$.

Therefore, we can compare the series $\sum_{n=1}^{\infty} a_n$ with the geometric series $\sum_{n=1}^{\infty} L^n$. In this way, we get parts (i) and (ii) in the test.

For part (iii), we can also look at the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ and $\sum_{n=1}^{\infty} \frac{1}{n}$, where $L = 1$ for both series. However, now one series is convergent but the other one is divergent.

Example 6. Determine the absolute convergence or divergence of the following series.

$$(1) \quad \sum_{n=1}^{\infty} (-1)^n \left(\frac{2n+1}{3n+2} \right)^n.$$

Now $\sqrt[n]{|a_n|} = \frac{2n+1}{3n+2} \rightarrow \frac{2}{3} < 1$. Therefore, the series is absolutely convergent.

$$(2) \quad \sum_{n=1}^{\infty} \left(1 + \frac{1}{n} \right)^{n^2}.$$

Now $\sqrt[n]{|a_n|} = \left(1 + \frac{1}{n} \right)^n \rightarrow e > 1$. Therefore, the series is divergent.

The relation between the Ratio Test and the Root Test.

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L \text{ exists} \quad \implies \quad \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L \text{ exists}.$$

The converse is not true. For example, let $a_n = \begin{cases} \frac{1}{2^n} & \text{if } n \text{ is odd} \\ \frac{1}{2^{n-2}} & \text{if } n \text{ is even} \end{cases}.$

Then we have $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \frac{1}{2}$, but $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ DNE.

Section 11.7: Some Strategies for Testing Series

The basic/standard series include geometric series, telescoping series, p-series, and the alternating series.

Test convergence/divergence of $\sum_{n=1}^{\infty} a_n$ by the following order:

- 1) If $a_n \not\rightarrow 0$, then the series is divergent by the Test for Divergence.
- 2) Check whether $\sum_{n=1}^{\infty} a_n$ is a linear combination of some basic series.
- 3) Consider $\sum_{n=1}^{\infty} |a_n|$ and use tests for positive series.
 - (i) Check whether $\sum_{n=1}^{\infty} |a_n|$ is a geometric series, a telescoping series, or a p -series.
 - (ii) Check whether there is a natural comparison with a geometric series or a p -series.
 - (iii) Use the Ratio Test if a_n contains $n!$ or an algebraic function of n .
 - (iv) Use the Root Test if a_n has the form $(b_n)^n$.
 - (v) If $a_n = f(n)$ for some decreasing continuous function $f \geq 0$ and it is easy to find the improper integral $\int_a^{\infty} f(x) dx$, then use the Integral Test.